HEADLIGHT AUTO LEVELING SYSTEM

id091800010900

Outline

- The headlight auto leveling system automatically adjusts the headlight LO optical axis in response to changes in load and passenger conditions to prevent blinding of oncoming vehicles from headlight glare and to assure a range of visibility.

Function

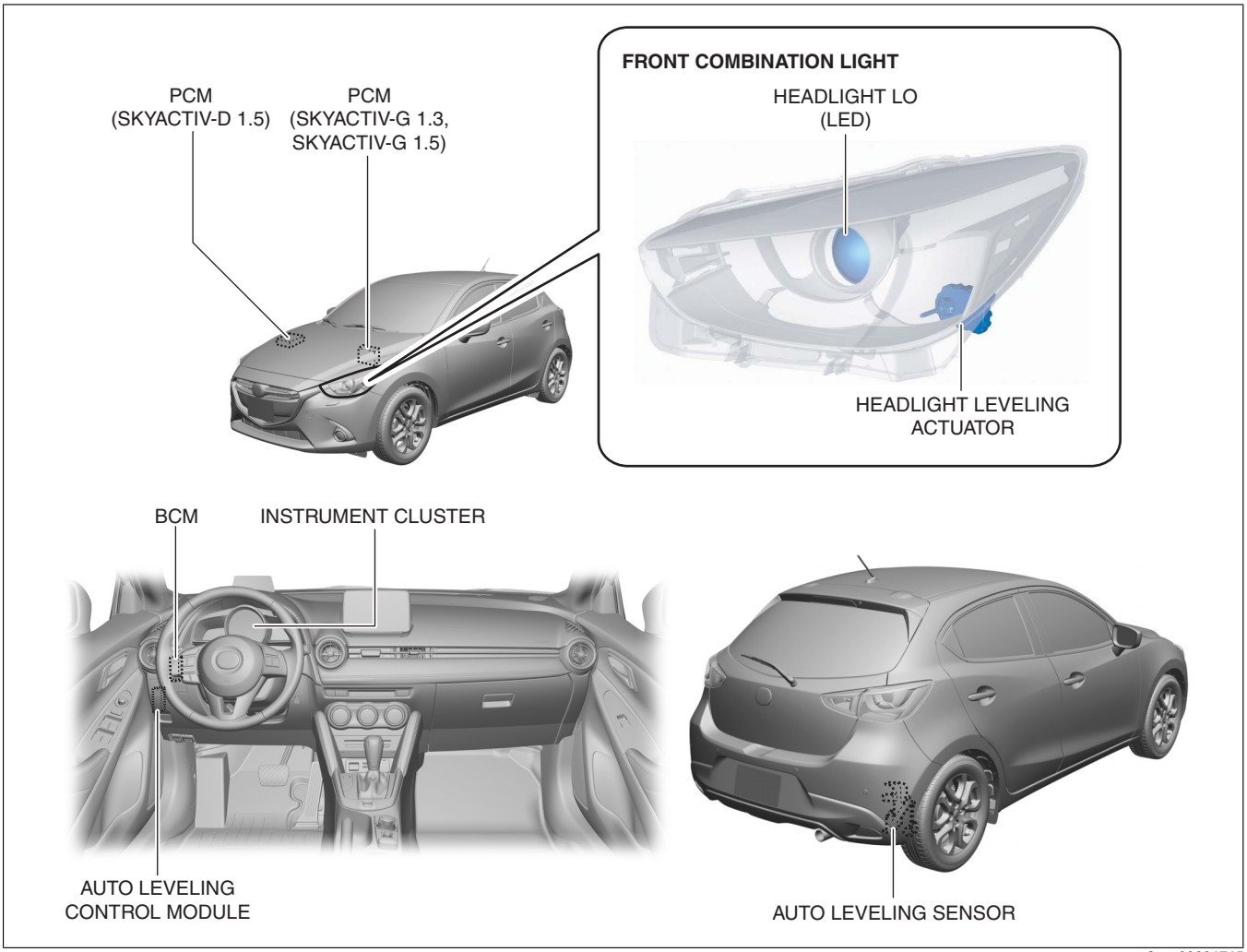
- The headlight auto leveling system is controlled by the auto leveling control module.

Optical axis adjustment function

- If there is a vehicle height change with the ignition switched ON (engine on or off) and the headlights LO turned on, the auto leveling control module automatically adjusts the headlight LO optical axis.
- The auto leveling control module controls the optical axis adjustment function based on the following signals.

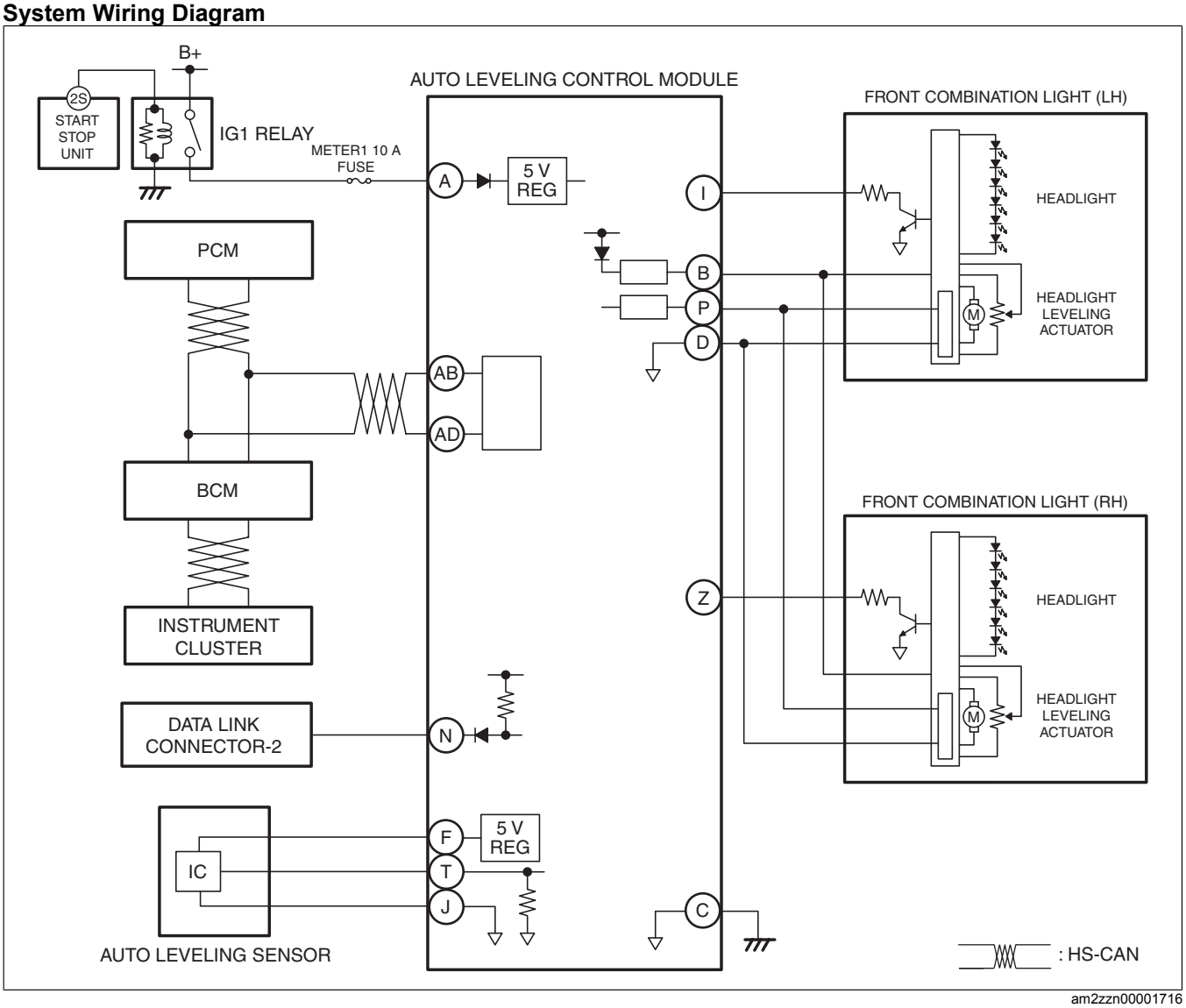
Signal name	Sending module/part name	Communication method
Vehicle speed signal	PCM	HS-CAN
Ignition switch position signal	Instrument cluster	
Headlight LO on condition signal	BCM	

Structural View



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System Wiring Diagram



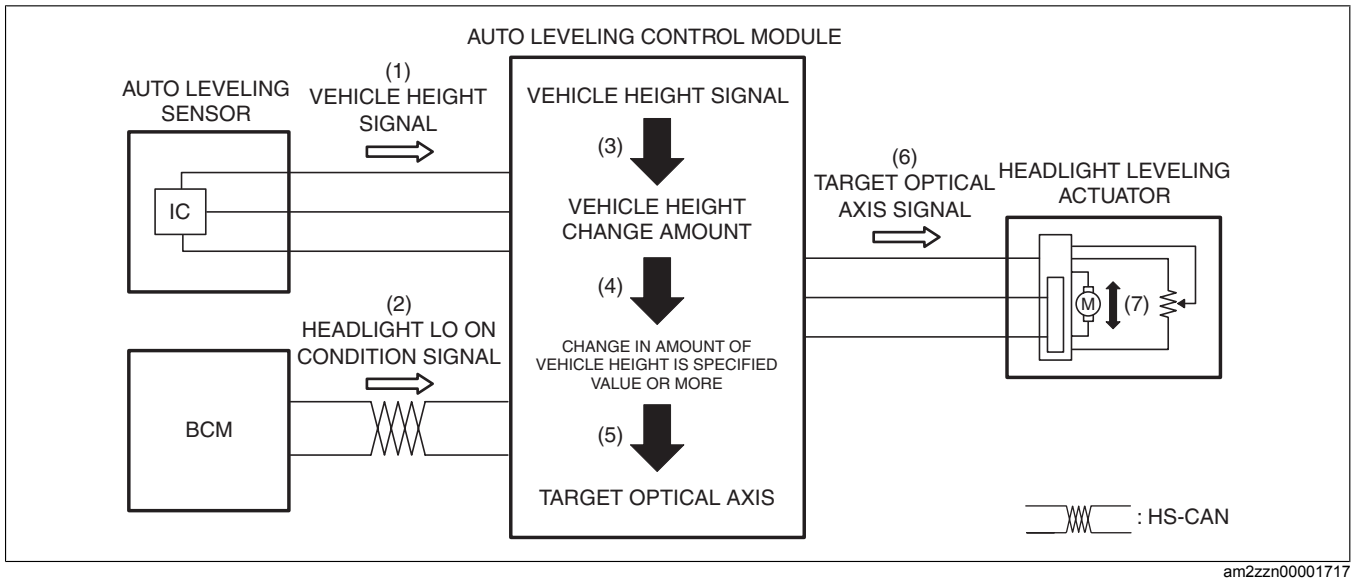
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Operation
Vehicle height changes while vehicle is stopped
Operation conditions

- Operates when all of the following conditions are met:
  - — Vehicle speed is **0 km/h {0 mph}** (**0 km/h {0 mph}** vehicle speed signal is received)
  - — Ignition is switched ON (engine off or on) (ignition switch status signal (ON) is received)
  - — Headlights LO on (headlight LO on condition signal reception)
  - — Detects change in vehicle height amount* at specified value or more

Operation

1. When the ignition is switched ON (engine off or on), the auto leveling sensor sends (1) a vehicle height signal to the auto leveling control module.
2. When the auto leveling control module receives (2) a headlight LO on condition signal from the BCM via CAN communication, it calculates (3) the amount of change in the vehicle height* based on the vehicle height signal.
3. When the auto leveling control module detects (4) the amount of change in the vehicle height* at the specified value or more, it calculates the target optical axis (5) based on the calculated amount of change in the vehicle height*.
4. The auto leveling control module sends (6) the calculated target optical axis signal to the headlight leveling actuator.
5. When the target optical axis signal is received, the headlight leveling actuator drives (7) the motor.



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* : The change in the amount of vehicle height is the calculated **1 s** average value of the vehicle height signal which the auto leveling control module detects every **0.1 s**.

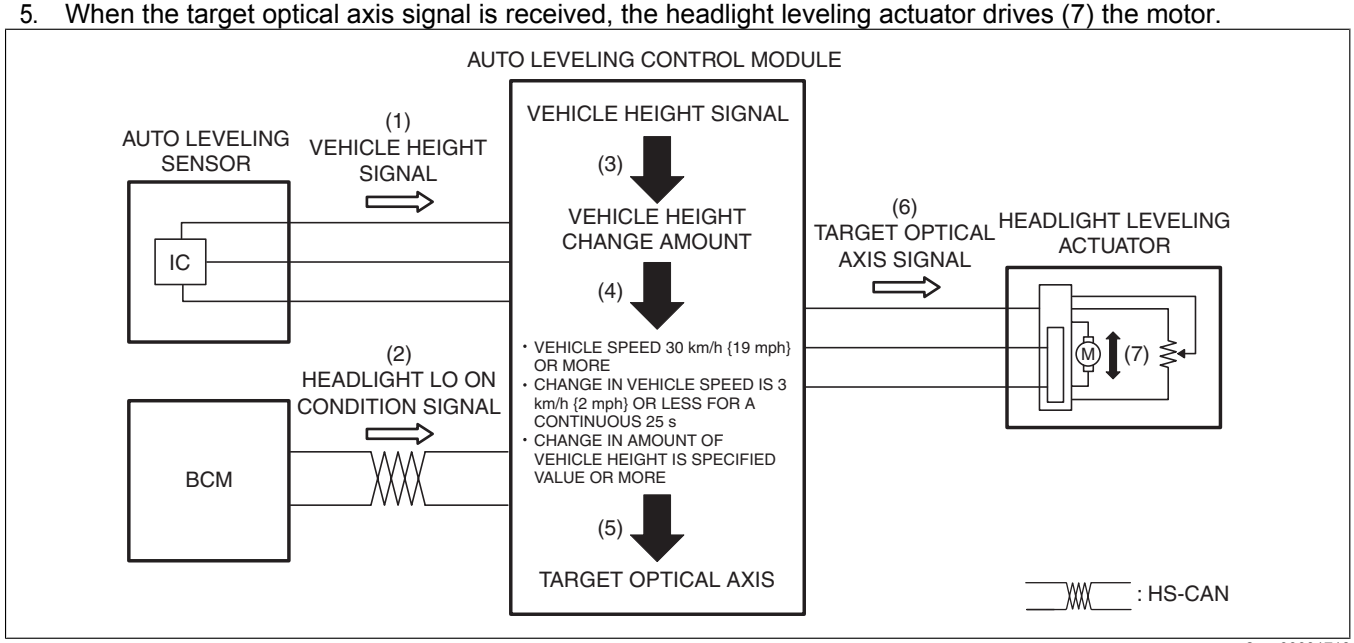
Vehicle height changes while vehicle is moving

Operation conditions

- Operates when all of the following conditions are met:
  - — Vehicle speed is **30 km/h {19 mph} or more** (vehicle speed signal of **30 km/h {19 mph} or more** is received)
  - — Headlight LO on (headlight LO on condition signal reception)
  - — Detects change in vehicle speed of **3 km/h {2 mph} or less** for continuous period of **25 s**
  - — Detects change in vehicle height amount* at specified value or more

Operation

1. The auto leveling sensor sends (1) a vehicle height signal to the auto leveling control module while the vehicle is being driven.
2. When the auto leveling control module receives (2) a headlight LO on condition signal from the BCM via CAN communication, it calculates (3) the amount of change in the vehicle height* based on the vehicle height signal.
3. When the auto leveling control module detects (4) the vehicle speed of **30 km/h {19 mph} or more**, change in vehicle speed of **3 km/h {2 mph} or less** for a continuous **25 s**, and the change in the amount of vehicle height* at the specified value or more, it calculates (5) the target optical axis based on the calculated change in the amount of vehicle height*.
4. The auto leveling control module sends (6) the calculated target optical axis signal to the headlight leveling actuator.
5. When the target optical axis signal is received, the headlight leveling actuator drives (7) the motor.



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* : The change in the amount of vehicle height is the calculated **1 s** average value of the vehicle height signal which the auto leveling control module detects every **0.1 s**.